

Charotar University of Science and Technology
Education Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY/ MICROBIOLOGY)
SEMESTER I EXAMINATION (BACKLOG)
BT701/MI701: GENERAL MICROBIOLOGY

Date: 03.05.2010

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.

PART A

Total marks: 20

Q1. Choose the correct option and put $\sqrt$ mark in front of it:

1. Heat sensitive media components are sterilized by
(a) autoclaving (b) hot air oven (c) filtration (d) none of the above
2. In the following staining technique the cells appear as lighter bodies against dark back- ground
(a) gram's (b) acid fast (c) endospore (d) negative
3. The major difference between the phase contrast and DIC microscope is the
(a) light source (b) path of light (c) λ of light (d) none of the above
4. The following is **NOT** the distinguishing feature of a fungal cell
(a) DNA associated with histones (b) 18S rRNA
(c) a single circular chromosome (d) all of the above
5. According to "hydrogen hypothesis" the endosymbiont was:
(a) anaerobic producing H₂ and CO₂ (b) aerobic utilizing H₂ and CO₂
(c) Photosynthetic (d) none of the above
6. PCR amplification **IS NOT** required for the following of microorganisms:
(a) cultivation (b) microscopic examination (c) axenic cultures (d) all of the above
7. The sequences that read the same on the complementary strand are called
(a) complementary (b) palindromes (c) repetitive (d) none of the above
8. The increase in sequence dissimilarity after divergence may be due to
(a) parallel substitution (b) back substitution
(c) convergent substitution (d) none of the above
- 9 For determining the pattern of spontaneous mutations under selective neutrality one should examine:
(a) genes coding for RNA (b) promoter region (c) exons (d) pseudogenes

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10. MLST involves sequence comparison of fragments of six to seven housekeeping genes from:
(a) type species (b) different species of same genus (c) different strain of same organism (d) all of the above
11. After gene duplication the size of DNA
(a) decreases (b) increases (c) remains the same (d) all of the above
12. Genetic diversity of microbial community IS NOT assessed by measuring the following of the DNA from the entire community:
(a) homogeneity (b) heterogeneity (c) G + C content (d) none of the above
13. "Trisporic acid suppressed asexual reproduction" is found in the following family
(a) *Mucorales* (b) *Entomorphthorales* (c) *Zoophagales* (d) none of the above
14. In *Chytridiomycota* the flagellum structure present is
(a) $9(3) + 0$ (b) $9(2) + 1$ (c) $9(2) + 0$ (d) $9(3) + 1$
15. Clamp connection is found in the following class of fungi
(a) *Ascomycota* (b) *Deuteromycota* (c) *Basidiomycota* (d) *Zygomycota*
16. Para sexuality is found in the following class of fungi
(a) *Zygomycota* (b) *Basidiomycota* (c) *Ascomycota* (d) none of the above
17. The following technique is used for the determination of microbial growth
(a) cell forming units (b) microbial dry weight
(c) spectrophotometry (d) all of the above
18. Mean doubling time of an organism growing as a batch culture is equivalent to
(a) doubling time (b) time required for DNA replication
(c) Generation time (d) all of the above
19. Exopolysaccharides produced by some microorganisms help in the formation of
(a) biofilms (b) colonies (c) lichen (d) all of the above
20. The turbidostat operates best at the following dilution rates
(a) high (b) low (c) intermediate (d) all of the above

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2. Write answers for Section I and Section II of Part B in separate answer sheets.

Part B

Total Marks: 50

Section: I

Q.1.A. Differentiate between ANY TWO of the following: (06)
(i) Slipped strand mispairing and unequal crossingover
(ii) Transposition and Retroposition
(iii) Classical Taxonomy and Chemical Taxonomy

Q.1.B. Describe ANY ONE of the following (03)
(i) The experiment carried out to prove that penicillin resistance in *N. gonorrhoeae* is due to horizontal gene transfer.
(ii) Shuffling of domain is more likely to succeed than shuffling of exon

Q.1.C. Justify: Retrosequences are also called "processed retrosequences". (01)

Q.2.A. Explain briefly ANY TWO of the following: (06)
(i) Evolutionary distance
(ii) Need for axenic culture to study microbial diversity in an ecosystem
(iii) Molecular chronometers

Q.2.B. Justify ANY ONE of the following: (04)
(i) Species diversity is low in physically controlled ecosystem than that in biologically controlled ecosystem.
(ii) Presence of plasmid that is responsible for a phenotypic traits may lead to over-estimation of relatedness of one organism to another.

Section:II

Q.3.A Discuss chemical disinfectants . (05)

Q.3.B Write short notes on any two of the following: (05)
(i) Mechanism of gram's staining
(ii) Principle of Light microscope
(iii) Sterilizing gases

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Q.4.A. Describe briefly ANY TWO of the following:

(06)

- (i) The life cycle of chlamydomonas
- (ii) Parasexual cycle
- (iii) Regulation of heterocyst formation in cyanobacteria

Q.4.B. Differentiate and compare between ANY ONE of the following

(04)

- (i) Ascomycota and Basidiomycota
- (ii) Zygomycota and Chytridiomycota

Q.5.A. Differentiate between ANY TWO the following

(04)

- (i) Heterotrophs and autotrophs
- (ii) Batch and continuous growth
- (iii) Enrichment and enriched media

Q.5.B. Define ANY SIX of the following:

(06)

- (i) Chemostat
- (ii) Turbidostat
- (iii) Defined media
- (iv) Selective Media
- (v) Differential media
- (vi) Growth
- (vii) Generation time
- (viii) Exponential growth